

TASK 1 — PROCESS DOCUMENTATION

Business Sales Performance Analysis | How I Completed the Task | Maincrafts Technology Internship
Harsh Parmar | Intern ID: MT5535 | Domain: Business Analytics (Excel + Power BI) | June 2026

1. TASK OBJECTIVE

The objective of Task 1 was to act as a **Business Analyst** and convert raw sales data into meaningful business insights using **Microsoft Excel and Power BI Desktop**. The task covered four core areas: data cleaning, KPI identification, dashboard creation, and business insight generation — mirroring real analytics work done in consulting firms and corporate teams.

2. DATASET OVERVIEW

Since no dataset was provided, I created a **realistic 150-row sales dataset** with the following structure:

Column	Data Type	Description	Sample Value
Order_ID	Text	Unique order identifier	ORD-4521
Order_Date	Date	Transaction date	01-Jun-2024
Region	Text	North/South/East/West	South
Product	Text	Product name	Laptop
Category	Text	Product category	Electronics
Sales_Amount	Number	Sale value in Rs.	45,200
Profit	Number	Profit earned in Rs.	6,328
Quantity	Integer	Units sold	3

The dataset covered **January 2024 to December 2024** with 4 regions, 5 categories, and 25 products. Intentional data quality issues were embedded — 5 duplicate rows and 5 blank Sales_Amount values — to simulate real-world dirty data.

3. PART A — EXCEL: DATA PREPARATION & ANALYSIS

Step 1: Data Import & Table Formatting

Opened the dataset in Microsoft Excel and converted the range into a formal **Excel Table (Insert → Table)** to enable structured referencing, auto-filtering, and dynamic range expansion for pivot tables.

Step 2: Data Cleaning

A

Remove Duplicate Rows

Used Data tab → Remove Duplicates selecting all 8 columns. This identified and removed 5 exact duplicate rows, reducing the dataset from 155 rows to 150 clean records.

B

Handle Blank Sales_Amount — Profit Margin Imputation Method

This was the most critical cleaning step. Instead of filling blanks with zero (which would distort KPIs) or a generic column average (which ignores product context), a Profit Margin-Based Imputation method was applied.

The mathematical logic behind this approach:

$\text{Profit} = \text{Sales_Amount} \times \text{Profit_Margin_Rate}$ So: $\text{Sales_Amount} = \text{Profit} \div \text{Profit_Margin_Rate}$ And: $\text{Profit_Margin_Rate} = \text{Avg}(\text{Profit of same product}) \div \text{Avg}(\text{Sales of same product})$

The Excel formula applied in helper column I (Cleaned_Sales):

`=IF(F2="", G2 / (AVERAGEIF($D$2:$D$156, D2, $G$2:$G$156) / AVERAGEIF($D$2:$D$156, D2, $F$2:$F$156)), F2)`

Five records with missing Sales_Amount were recovered using this method. For each affected record, the Sales_Amount was mathematically derived by dividing the known Profit by the average Profit Margin Rate of the same product — ensuring **product-specific accuracy** over generic mean substitution. This approach is used in real-world data science and analytics workflows under the technique known as **Group Mean Imputation**.

C

Date Formatting

Selected the Order_Date column and applied DD-MMM-YYYY format via Format Cells to ensure consistent date parsing for pivot table grouping by month.

Step 3: Calculated Columns

Two new columns were added to enrich the dataset for analysis:

Column I — Total_Sales = Sales_Amount × Quantity (e.g. =F2*H2) Column J — Profit_Margin% = (Profit ÷ Sales_Amount) × 100 (e.g. =IF(F2=0,0,(G2/F2)*100))

The IF condition in Profit_Margin% prevents a **#DIV/0!** error for any zero-value sales rows.

Step 4: Pivot Tables

Pivot Table	Row Field	Value Field	Sheet Name
Sales by Region	Region	Sum of Total_Sales	PT_Region
Sales by Product	Product	Sum of Total_Sales	PT_Product
Monthly Sales Trend	Order_Date (Month)	Sum of Total_Sales	PT_Monthly

4. PART B — POWER BI: DASHBOARD CREATION

Step 1: Data Import

Imported the cleaned Excel file into Power BI Desktop via **Home** → **Get Data** → **Excel Workbook**. Selected the Raw Sales Data sheet and loaded it into the Power BI data model.

Step 2: DAX Measures Created

Total Sales = SUM('Raw Sales Data'[Sales_Amount]) Total Profit = SUM('Raw Sales Data'[Profit])

These measures were created by right-clicking the table in the Fields panel → New Measure. DAX (Data Analysis Expressions) measures are preferred over column aggregations as they respond dynamically to slicer filters.

Step 3: Visuals Built

Visual	Type	Fields Used	Purpose
KPI Card 1	Card Visual	Total Sales	Show overall revenue at a glance
KPI Card 2	Card Visual	Total Profit	Show overall profitability
Top Products	Clustered Bar Chart	Product + Total Sales	Compare top 5 products by revenue
Monthly Trend	Line Chart	Order_Date + Total Sales	Visualize month-on-month sales pattern
Region Sales	Pie Chart	Region + Total Sales	Show regional revenue contribution %

Note: Map chart was not used for Region-wise sales because the dataset uses directional labels (North/South/East/West) rather than real geographic place names. **Pie Chart was chosen as the appropriate substitute.**

Step 4: Slicers Added

Three slicers were added to make the dashboard interactive and decision-ready: **Date, Region, and Product Category**. These allow management to filter all visuals simultaneously and explore the data from any perspective without changing the underlying report.

5. EXPLORATORY DATA ANALYSIS (EDA) SUMMARY

EDA was performed to understand the structure, quality, and distribution of the sales data before building any visuals. Key findings:

Data Shape	150 rows × 8 columns after cleaning. Original file had 155 rows with 5 duplicates.
Missing Values	5 blank Sales_Amount values recovered using Profit Margin-Based Group Imputation.
Date Range	Full calendar year: January 2024 to December 2024 — enabling 12-month trend analysis.
Category Distribution	5 categories with Electronics heavily dominating at 69.2% of total sales volume.
Regional Distribution	4 regions relatively balanced — South (28.2%) to West (20.6%) — a 7.6% spread.
Outliers	No extreme outliers detected. Product-level pricing was consistent within category ranges.
Profit Margin Range	13.6% (Electronics) to 37.3% (Clothing) — significant variation across categories.
Seasonality	Clear mid-year dip (June) and Q4 peak (November) identified in monthly trend analysis.

6. FINAL DELIVERABLES SUBMITTED

1	Cleaned Excel File Raw data cleaned, calculated columns added, 3 pivot tables built (Sales by Region, Sales by Product, Monthly Trend).
2	Power BI Dashboard (.pbix) Interactive dashboard with 2 KPI cards, bar chart, line chart, pie chart, and 3 slicers for Date, Region, and Category.
3	Insight Report (PDF/Word) One-page professional document with 5 key insights in story format and 5 actionable business recommendations with estimated revenue impact.
4	Process Documentation

This document — explaining the complete methodology, EDA approach, formulas used, and decisions made throughout the task.

Harsh Parmar

Business Analytics Intern | MT5535 | Maincrafts Technology

Task 1 — Process & EDA Documentation

Business Analytics (Excel + Power BI) | June 2026